

Data Visualization using Python (Matplotlib, Seaborn & Dataset Analysis)

1. Introduction to Data Visualization

Data visualization means presenting data in graphical format such as charts, graphs, plots, and dashboards. Visualization helps users understand patterns, trends, comparisons, and relationships in data. Why Data Visualization is Important: • Makes complex data easy to understand • Helps in better decision making • Detects trends and patterns quickly • Useful in business analytics, machine learning, AI, healthcare, finance, and education Real-life examples: • YouTube analytics graphs • IPL score analysis • Weather forecast charts • E-commerce sales dashboards

2. Libraries Used

Matplotlib: Matplotlib is a Python library used to create basic charts such as line plots, bar graphs, pie charts, and histograms. Seaborn: Seaborn is built on top of Matplotlib and is used for statistical data visualization. It provides beautiful and attractive plots with less coding. Pandas: Pandas is used for data handling, cleaning, and analysis.

3. Installing Libraries

Install the required libraries using pip:

```
pip install matplotlib
pip install seaborn
pip install pandas
```

4. Importing Libraries

Before creating visualizations, import the necessary libraries.

```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
```

5. Line Plot

A line plot is used to show trends over time. It connects data points using lines. Real-world use: • Monthly sales growth • Temperature changes • Stock market analysis Explanation of Code: • x contains x-axis values • y contains y-axis values • plt.plot() creates the line graph • plt.title() adds title • plt.xlabel() adds x-axis label • plt.ylabel() adds y-axis label • plt.show() displays graph

```
import matplotlib.pyplot as plt

x = [1,2,3,4,5]
y = [10,20,25,30,40]

plt.plot(x, y, marker='o')
plt.title("Line Plot")
plt.xlabel("X-axis")
plt.ylabel("Y-axis")
plt.show()
```

6. Bar Chart

A bar chart is used to compare categories. Real-world use: • Comparing student marks • Product sales comparison • Population comparison Explanation: • Each bar represents a category • Height of bar represents value

```
students = ["A", "B", "C", "D"]
marks = [80, 65, 90, 75]

plt.bar(students, marks)
plt.title("Student Marks")
plt.xlabel("Students")
plt.ylabel("Marks")
plt.show()
```

7. Histogram

A histogram shows frequency distribution of numerical data. Real-world use: • Age distribution • Salary distribution • Exam marks analysis Explanation: • bins divide data into intervals • Histogram helps identify data spread

```
data = [1,2,2,3,3,3,4,4,5]

plt.hist(data, bins=5)
plt.title("Histogram")
plt.show()
```

8. Scatter Plot

A scatter plot shows relationships between two variables. Real-world use: • Height vs weight • Study hours vs marks • Advertising cost vs sales Explanation: • Each point represents one observation • Helps identify correlation

```
x = [1,2,3,4,5]
y = [2,4,5,4,5]

plt.scatter(x, y)
plt.title("Scatter Plot")
plt.show()
```

9. Pie Chart

A pie chart shows percentage contribution of categories. Real-world use: • Market share • Election results • Expense distribution Explanation: • labels represent category names • autopct displays percentages

```
subjects = ["Python", "Java", "C++", "AI"]
students = [40, 30, 20, 10]

plt.pie(students, labels=subjects, autopct='%1.1f%%')
plt.show()
```

10. Seaborn Visualizations

Seaborn provides attractive and advanced statistical plots. Advantages: • Better design • Less coding • Statistical analysis support

```
import seaborn as sns

tips = sns.load_dataset("tips")

sns.scatterplot(x="total_bill", y="tip", data=tips)
plt.show()
```

11. Box Plot

A box plot helps detect: • Outliers • Median • Quartiles Real-world use: • Salary analysis • Exam result analysis

```
sns.boxplot(x="day", y="total_bill", data=tips)
plt.show()
```

12. Heatmap

A heatmap displays correlation using colors. Correlation Values: • +1 = Strong positive correlation • 0 = No correlation • -1 = Strong negative correlation Heatmaps are widely used in machine learning and data science.

```
corr = tips.corr(numeric_only=True)

sns.heatmap(corr, annot=True)
plt.show()
```

13. Pairplot

Pairplot creates multiple plots together to show relationships between all columns. Useful for: • Dataset exploration • Feature analysis • Machine learning preprocessing

```
sns.pairplot(tips)
plt.show()
```

14. Hands-on Dataset Analysis

In this section, students will analyze a simple student performance dataset. Tasks: • Create DataFrame • Analyze dataset • Create charts • Find topper student • Understand correlations

```
import pandas as pd

data = {
    "Name": ["A", "B", "C", "D", "E"],
    "Math": [90, 78, 85, 60, 95],
    "Science": [88, 67, 80, 70, 98],
    "English": [75, 80, 85, 72, 90]
}

df = pd.DataFrame(data)

print(df)
```

Explanation: • Dictionary stores data • DataFrame converts data into table format • print(df) displays dataset

```
print(df.head())
```

```
print(df.describe())  
print(df.info())
```

head(): Displays first 5 rows. describe(): Shows statistical summary like mean, max, min. info(): Displays column names and datatype information.

15. Final Classroom Activity

Final Classroom Activity: Students should create a mini dashboard using: • Line chart • Bar chart • Histogram • Heatmap
Students should also identify: • Highest scorer • Average marks • Correlation between subjects

16. Interview Questions

1. What is data visualization? 2. Difference between Matplotlib and Seaborn? 3. What is a histogram? 4. What is correlation? 5. What is a heatmap? 6. Why is visualization important in Machine Learning?

17. Homework

1. Create IPL Data Analysis graphs. 2. Analyze Netflix dataset. 3. Create student result dashboard. 4. Compare Matplotlib and Seaborn outputs.